

INDUSTRY PROJECT REPORT

Title : Employee Attrition Analysis using Power BI

Name of the Company : TCS iON

Name of the Institute : Gurunanak Institute of Technology

Project Duration : 18 - 04 - 2026 to 22 - 04 - 2026

Total Effort (hrs.) : 45 Hours

Submitted By : Priya Yadav

Project Environment

- Windows OS
- Jupyter Notebook
- Power BI
- VS

Tools Used

- Python (Pandas, NumPy, Matplotlib, Seaborn)
- Power BI
- Excel / CSV
- SQL

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1. Acknowledgement

I would like to thank my institute and mentors for their support and guidance throughout this project.

2. Objective and Scope

The objective of this project is to analyze employee attrition data and identify key factors affecting employee turnover. The scope includes data cleaning, analysis, and dashboard creation.

3. Problem Statement

Employee attrition is a major issue for organizations. The goal is to identify patterns and insights to reduce employee turnover.

4. Existing Approaches

Traditional methods rely on manual analysis and reports, which are time-consuming and less interactive.

5. Approach / Methodology

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Visualization using Power BI
- KPI and dashboard creation
- Forecasting analysis

6. Workflow

- Load dataset
- Clean data (remove nulls, duplicates)
- Perform EDA
- Create visualizations
- Build dashboard

7. Assumptions

- Data is accurate and reliable
- Attrition values are correctly defined (Yes/No)

8. Implementation

- Cleaned dataset (removed null values and duplicates)
- Standardized categorical data
- Created charts:
 - Attrition by Department
 - Attrition by Job Role
 - Gender Distribution
 - Experience vs Salary
- Built KPIs:
 - Total Employees
 - Attrition Count
 - Attrition Rate
 - Average Salary
 - Average Experience

9. Solution Design

A Power BI dashboard was designed with:

- KPI cards
- Drill-down charts
- Filters (Department, Job Role, Gender)
- Forecasting visuals

10. Challenges & Opportunities

- Data inconsistency (MALE/male, YES/yes)
- Handling missing values

11. Reflections on the Project

This project helped in understanding data cleaning, visualization, and dashboard building.

12. Recommendations

- Improve retention for new employees
- Provide career growth opportunities
- Offer competitive salaries

13. Outcome / Conclusion

The project successfully identified key factors affecting attrition and provided insights for decision-making.

14. Enhancement Scope

- Add machine learning models for prediction
- Include more features like job satisfaction
- Improve dashboard design

15. Link to Code

Git Hub link : <https://github.com/priyayadav2004>

16. Research Questions

- What factors affect attrition?
- Which department has highest attrition?
- Does salary impact attrition?

17. References

- Power BI Documentation
- Python Libraries Documentation